

AI Engineering Assignment: Talk With Your Data

Estimated time: 4-8 hours development time. We care more about depth of reflection and what you learned than feature count.

The task

Build a small "**talk with your data**" app: a user asks a question in natural language, the app figures out the answer from the dataset (typically by generating and running SQL), and explains the result.

Dataset: [Olist Brazilian E-Commerce](#). Use **at least these three tables**: `orders`, `order_reviews`, and `order_customers`. Add more tables if it serves your app, your call.

Core requirements

- Load and model the data sensibly (your choice of storage; SQLite, DuckDB, Postgres, whatever)
- A working natural-language → answer loop (text-to-SQL is the expected baseline; if you choose a different approach, justify why)
- The app should **handle ambiguous or broken questions gracefully**; try to break it yourself and note what happens and explain why.

Bonus directions (pick what excites you, don't do all)

- A minimal frontend or interesting visualizations
- A trained model for a prediction task (review score, late delivery, churn, ...)
- Hybrid SQL + semantic search over the free-text reviews
- A small eval set (5–10 questions with expected behavior) showing how your app performs
- Anything else you find interesting, surprise us

What we're evaluating

We care about **process and learning** at least as much as the final artifact:

- **How you built it** - fully manual, fully AI-assisted, hybrid? Which tools, which workflow, how did you manage context, did you use planning/subagents? *There are better and worse ways to code with AI, show us you've thought about it, try something new and explain what you learned from this.*
- **Mastery of the codebase and concept** – You don't need to have read every line, but you should be the conceptual owner: expect us to point at parts of the app and ask you to explain the details, control flow, and design choices.
- **New skills** - use **at least one** tool, library, technique, or AI engineering practice you haven't used before. Ideally more. What did you learn from it?

- **Technical highlight** - one or two things in the code, model, or feature you're proud of.
- **Creativity & framing** - what story does your app tell? What's interesting about it?
- **Reflection** - what broke or didn't work, what you'd do next with more time, what surprised you.

What we do *not* care about

Deployment, auth, polished UI, exhaustive documentation. Don't over-engineer.

Practical notes

- **Vibecoding is encouraged.** Use AI assistance freely and have fun with it but don't outsource your brain.
- **No agentic coding tool yet?** Try the free tier of [Google Antigravity](#). Cursor, Claude Code, Codex, OpenCode etc. are all fine too.
- **Need an LLM API key for the app itself?** We can provide a temporary Claude Sonnet key, just ask.

Deliverable & presentation

- A repo (GitHub) with (runnable) code and a short README
- You'll **present the app live** (~60 min): walk us through what it does, how you built it, your AI-assisted workflow, the technical highlight, and what you learned. No formal write-up or screen recording needed, but feel free to jot notes to structure your thoughts around the evaluation criteria above.

Have fun with it.